

DeltaSeek: Toward Active Perception in Evolving Construction Environments

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Abstract—Construction environments evolve continuously. Elements are installed, temporary objects appear and disappear, materials become occluded, and surfaces that were visible last week are inaccessible today. Robotic perception in such settings cannot always be treated as a static mapping problem in which observations are gathered first and interpreted afterward: large geometric and temporal change reduces correspondence between observations taken at different times, making registration and persistent scene understanding progressively harder. This motivates a more active formulation, in which a robot deliberately selects sensing configurations that supply the information needed to resolve the current state of the environment.

We present DeltaSeek, an initial framework toward active perception in evolving built environments. The broader objective is a robot that reasons about where to observe, how to observe, and ultimately when enough evidence has been gathered. As a first step we study a question underlying all three: how does sensing embodiment constrain the observations available to a mobile robot? We build an IFC-derived simulation benchmark and evaluate a Husky A300 with a UR5e under chassis-mounted and wrist-mounted RGB-D configurations, scoring candidate observations by geometric visibility and effort by drivable distance on the platform’s navigation costmap.

In a room-scale environment containing eight controlled scene changes spanning four observability conditions, an exhaustive evaluation over 240 admissible base poses shows that two changes cannot be observed by the chassis camera from any tested viewpoint, while the wrist camera resolves both. Changes constructed around occlusion and viewing incidence remain observable from the chassis but demand less accessible viewpoints and more travel: median drivable distance to first observation rises from 6.0 m with wrist-mounted sensing to 14.2 m with chassis-mounted sensing. These results separate two effects that matter for active perception — sensing capability, where embodiment decides whether an informative observation exists at all, and acquisition cost, where the same information is available but must be paid for in motion. We argue this distinction is a necessary foundation for robots that plan perception rather than passively accumulate data.

I. INTRODUCTION

A construction site is not a scene to be mapped once. Between two visits, walls are erected, services are hung, pallets are delivered and consumed, and formwork is struck. The same physical space presents a materially different surface geometry from week to week, and the parts that changed are precisely the parts a verification system cares about.

Most robotic capture pipelines in construction are nevertheless organised around passive accumulation: drive a coverage route, collect point clouds or images, register

them to the model, and difference them offline [1], [2], [3]. That is reasonable when the environment is nearly static, and it degrades when it is not. Registration relies on correspondence between what was seen before and what is seen now, and large change is exactly when that correspondence thins out: the information the robot most needs is what its pipeline is least equipped to acquire.

Active perception offers the alternative framing [4], [5]: treat sensor placement as a decision that resolves a specific uncertainty. Applied to an evolving building, this means asking where to look next to determine whether part of the model still describes reality — which requires answering, in sequence, where to observe, how to observe, and when enough evidence has been gathered.

This paper takes a deliberately narrow first step. Before a planner can choose among sensing configurations, we need to know what the robot’s embodiment makes available at all. A mobile manipulator can place a camera almost anywhere in its reachable workspace; a fixed chassis sensor cannot. That freedom is usually assumed valuable. We measure it.

Fig. 1 shows the setup and the effect it measures. Our contribution is threefold. First, a reproducible IFC-derived benchmark that clips a real Revit export into a room-scale scene and injects controlled changes with exact ground truth. Second, an observability taxonomy that classifies a change by why it is hard to see rather than by what moved. Third, an exhaustive sensor-configuration study whose central finding is a distinction we believe generalises: embodiment separates changes into those a sensor configuration cannot observe at all, and those it can observe only at greater cost.

II. RELATED WORK

As-built verification against a building model is well established, from dimensional compliance checking [1] to progress tracking [2], [3]. These systems consume data rather than plan it: the route is set by coverage or an operator, and the verification objective never influences where the sensor goes.

Active perception treats sensing as an action to be planned [4], [5]. Next-best-view methods maximise expected information for reconstruction [6], [8] or exploration [7], [9], usually under an unknown-space objective. Our setting differs: a strong prior exists, and the question is what has changed. Mounting the sensor on an arm enlarges the achievable viewpoint set, but how much that is worth is rarely quantified.

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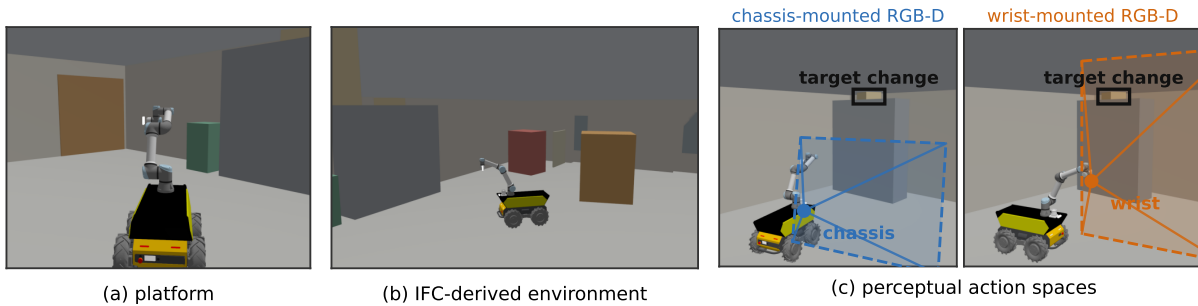


Fig. 1. The study, in simulation. (a) A Husky A300 with a UR5e, carrying a chassis RGB-D camera and a lidar on the deck and an identical RGB-D camera at the wrist. (b) Hall B, clipped from a real Revit IFC export, with eight injected changes. (c) Perceptual action spaces: each sub-panel uses the base yaw that aims its sensor at the same target, since the arm rotates the wrist camera 90° off the base heading. Each frustum is drawn to the target's depth, so its far face lies in the target's plane — the change falls outside the chassis frustum and inside the wrist one. Renders and frusta come from the camera poses the evaluation uses.

III. BENCHMARK

A. From IFC to a room-scale scene

The benchmark derives its geometry from a real Revit 2025 IFC2X3 [10] export of a single-storey building, $44.5 \times 19.0 \times 4.5$ m, comprising 133 products that convert without failure through IfcOpenShell [11]. Each product is approximated by a box oriented in its own placement frame. Orientation is not a refinement: the export sits on a site grid rotated 0.90° off the world axes, and a world-aligned box turns a 0.203 m wall into a 0.82 m one, a fourfold error on the dimension that decides whether the robot fits between them.

A storey is the wrong unit for studying observation, and is unusable directly — one wall entity runs the full 39 m of a facade. We clip a manifest to a rectangle, cutting in each element's own frame so the rotation survives. The result, Hall B, is 7.0×12.5 m with twelve modelled elements. Two fit-out units are added, since the export contains no furniture and both occlusion and height-limited observability need something to stand behind or on top of.

B. Observation model

Detection is geometric rather than image-based. An element's surface is sampled, and a sample counts when it falls inside the view frustum, faces the camera within a 75° incidence limit, and is unoccluded by another element. A change is observed when the visible surface fraction supporting it exceeds 0.05, which stands in for a detector's sensitivity. Each change type carries its own evidence: an unmodelled object is revealed by seeing it, a displaced element by seeing it in its new position or its modelled volume empty.

The model assumes perfect recognition and localisation, so it measures viewpoint quality rather than perception robustness — and it is fast enough to evaluate exhaustively, which rendering is not.

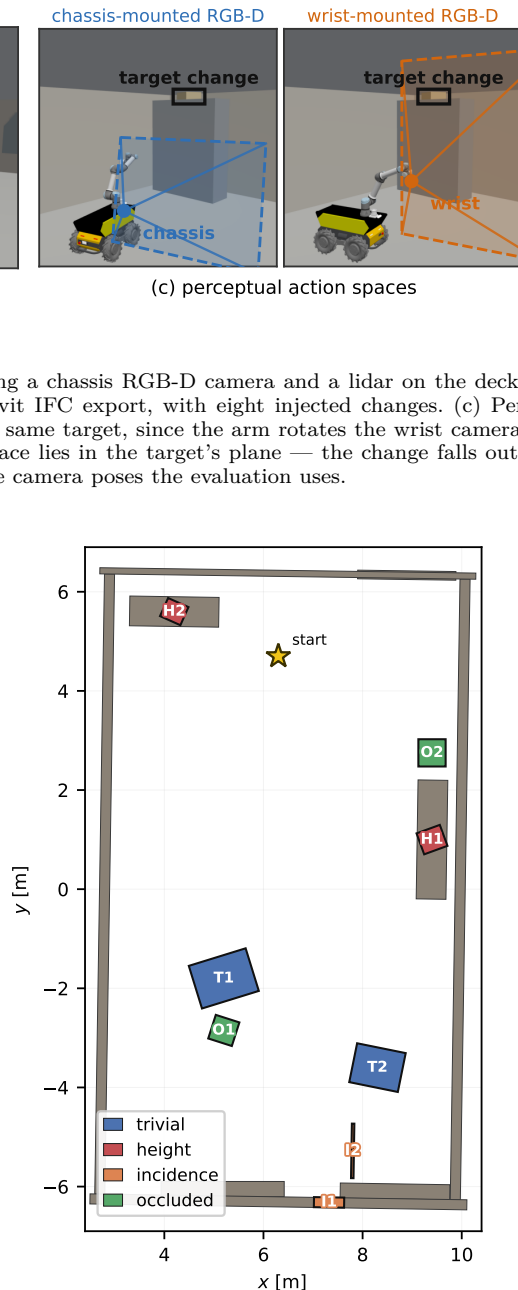


Fig. 2. Hall B, clipped from a real IFC export, with the eight injected changes coloured by observability class. Grey is the modelled shell and fit-out. The robot starts at the marked pose.

C. Effort

Inspection effort is drivable distance: Dijkstra search over an occupancy grid inflated by the platform's own Nav2 footprint (0.606 m circumscribed radius, 0.06 m resolution). This matters — on a storey-scale scene drivable distance averages $1.8\times$ the straight line and reaches $11\times$, and straight-line distance would let a robot pass through walls to reach a viewpoint.

TABLE I
Changes observed, by observability class

Class	chassis	wrist	both
trivial	2/2	2/2	2/2
height	0/2	2/2	2/2
incidence	1/2	2/2	2/2
occluded	2/2	2/2	2/2
all	5/8	8/8	8/8

D. Observability classes

A change’s type says what moved; it says nothing about whether a sensor can see it. We therefore label each change by why it is hard to observe:

- trivial — a large unmodelled object in open floor space;
- height — above the chassis camera’s sightline;
- incidence — resolvable only from an oblique angle;
- occluded — in another object’s shadow from most base poses.

Eight changes were hand-placed in Hall B, two per class, in a committed scenario rather than sampled, so that each is explainable from geometry and the study is exactly reproducible. The trivial pair is included as a control: a configuration that misses it is visibly broken.

IV. SENSING CONFIGURATIONS

The platform is a Clearpath Husky A300 with an AMP enclosure and a UR5e, generated from a single robot.yaml so that simulation and hardware share one description. It carries two RGB-D cameras of the same model, so the comparison isolates where a camera is mounted rather than how good it is:

The chassis camera is fixed on the enclosure deck at (0.460, 0.000, 0.418) m in the base frame, pitched 0.17 rad down, so its pose depends only on the base. The wrist camera is eye-in-hand, its pose obtained by forward kinematics over the generated description across five inspection postures; that chain is verified against the running simulation’s transform tree to machine precision.

Candidate base poses are laid on a 1.0 m grid of navigable free space inside the room, at four yaw bins 90° apart, giving 240 admissible base poses from 60 positions. Each combines with the five arm postures, so the wrist camera has 1200 distinct poses while the chassis camera has 240 — it does not move with the arm. An element is observed if any active sensor observes it.

V. RESULTS

A. Capability: what cannot be seen at all

Table I reports what each configuration observed within a 25 m budget, and Fig. 4(a) the underlying capability.

The headline is not 5/8. A single planned run conflates what a sensor can see with what one route reached, so we separate them by evaluating every admissible

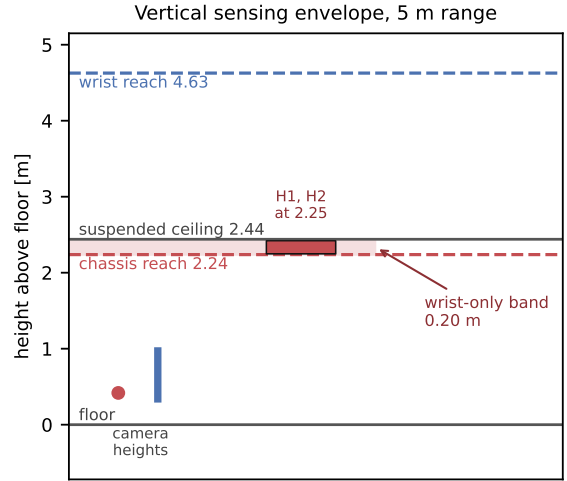


Fig. 3. Why exactly two changes are unreachable. At the 5 m range limit the highest point the chassis camera can place inside its frustum is 2.24 m; the wrist reaches 4.63 m. The ceiling begins at 2.44 m, leaving a wrist-only band 0.20 m tall, and the two height changes sit in it at 2.25 m. Both reaches are closed-form, and agree with a numeric sweep to 3 mm.

pose exhaustively. Two changes — both height — are observable from none of the 240 chassis poses, at zero visible fraction: a property of the embodiment. One further change the chassis-only run missed is observable from 14 of those 240 poses; the planner did not visit one inside the budget, which is routing, not evidence for the manipulator.

The distinction is not academic: repositioning the arm on the deck during development moved the chassis result from 6/8 to 5/8, which reads as the manipulator becoming more necessary and was entirely routing noise. The capability figure did not move.

Fig. 3 explains why only height produces a capability gap. For a camera of pitch p at height h , the highest world point inside its frustum is $h + r(\cos p \tan \frac{v}{2} - \sin p)$ at range r , with no yaw term: yaw rotates about the vertical axis and cannot change the vertical extent. At $r = 5$ m this gives 2.24 m for the chassis camera and 4.63 m for the wrist, against a ceiling at 2.44 m. The wrist-only band is therefore 0.20 m tall, and the two height changes were placed in it at 2.25 m.

B. Acquisition cost: what is merely expensive

The occluded and incidence pairs were designed to require the manipulator and did not. The crate behind the pallet stack is observable from 16 of 240 chassis poses — hidden from 93% of them, as intended — but a planner needs only one of the remaining sixteen. Where both configurations succeed the difference appears as distance, not detections (Fig. 4(b)): median drivable distance to first observation is 6.0 m for the wrist and 14.2 m for the chassis, a factor of 2.4.

Occlusion, the intuitive case for a manipulator, is in this environment a routing cost paid in metres, not a

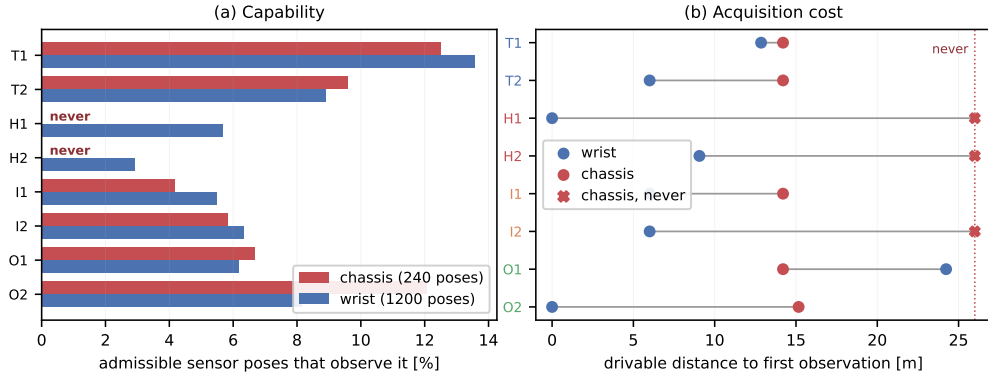


Fig. 4. (a) Capability: share of admissible sensor poses observing each change; two are observable from no chassis pose at all. (b) Acquisition cost: drivable distance to first observation under a 25 m budget.

sensing limit.

VI. DISCUSSION

Two effects fall out of these measurements, and an active-perception system must treat them differently. Sensing capability is binary and set by embodiment: no planning recovers a change outside every achievable frustum. Acquisition cost is continuous and set by the environment — the information exists, and the question is how much motion it takes. This is where viewpoint planning has leverage, and where the 6.0 m against 14.2 m gap lives. Conflating them is easy: a single run reports only “observed” or “not observed”, so an expensive change is indistinguishable from an impossible one.

The capability boundary is also a design variable. Hall B’s band is 0.20 m because the chassis camera sits at 0.418 m pitched 0.17 rad down; levelling it shrinks the band toward zero, and a taller space widens it. The useful output is not “the arm is worth two changes” but a method for locating that boundary before committing to a design.

VII. LIMITATIONS

The result is narrow: one room, one ceiling height, eight changes. The 0.21 m band is a property of this geometry, and a trend needs rooms of differing height. Detection is geometric — no detector, no renderer, no noise — so these numbers describe viewpoint quality, not whether a perception stack would recover the change. Elements are boxes, discarding the 48 opening entities in the source model, so Hall B is a sealed room the robot must be spawned inside. The height margin is also thin: the pair clears the chassis ceiling by 11.8 mm, so levelling or raising that camera would close the gap — which is the design-variable point made above, stated as a caution. Finally, the framework does not yet answer its third question: deciding when enough evidence has been gathered needs graded belief and a stopping rule, not binary detection.

VIII. CONCLUSION

DeltaSeek is a first step toward robots that plan perception in environments that will not hold still. Using an IFC-derived, reproducible benchmark we asked what a mobile manipulator’s embodiment buys. The answer is sharper and smaller than the usual assumption: two of eight changes lie outside the chassis camera’s reach entirely, and the rest cost roughly $2.4\times$ the travel without the arm. A robot that will one day decide when it has seen enough must first distinguish a view it cannot obtain from one it has not yet paid for.

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